

Gowtham Selvaraj

Backend Engineer • Java • Spring Boot • AWS • Microservices

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Professional Summary

Software Engineer with nearly **1 year of enterprise experience** at Cognizant, complemented by strong hands-on backend development through production-ready Java and Spring Boot projects. Skilled in building scalable RESTful APIs, secure backend systems, and cloud-native applications using **Java, Spring Boot, Spring Security, MySQL, Docker, and AWS**. Experienced in optimizing database performance, implementing clean architecture, and following Agile development practices. Passionate about designing high-performance, secure, and maintainable backend systems while continuously expanding expertise in microservices and cloud technologies.

Technical Skills

Languages	Java, SQL, Bash, HTML5, CSS3
Backend	Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Microservices Architecture, REST API Design
Frontend	React.js, Thymeleaf, Bootstrap, Tailwind CSS
Databases	MySQL, OracleSQL, Procedural SQL
Cloud & DevOps	AWS (EC2, S3, RDS, Lambda, SES), Docker, Git
Build Tools	Maven
Monitoring	Log4j, SLF4J, Prometheus, Grafana
Concepts	System Design, OOP, SOLID Principles, SDLC, Agile (Scrum), API Optimization, Cloud Security

Work Experience

Programmer Analyst Trainee

Nov 2024 – Present

Cognizant Technology Solutions (Client: FIS)

Kolkata, India

- Supporting the **FIS Account Origination** platform used by banking clients to manage customer onboarding and account creation workflows.
- Implemented client-requested business workflow changes using the Authority and Configuration Tool (ACT), configuring forms, validation rules, business logic, and process flows based on functional requirements.
- Analyzed production support tickets, investigated reported issues, and delivered configuration fixes while ensuring minimal impact on existing banking workflows.
- Collaborated with business analysts, project managers, and cross-functional teams to understand change requests, validate requirements, and deploy configuration updates during scheduled production releases.
- Participated in Agile Scrum ceremonies including sprint planning, daily stand-ups, backlog discussions, and release activities within an enterprise banking environment.
- Continuously strengthened backend engineering expertise outside project work by building production-ready applications using Java, Spring Boot, Spring Security, MySQL, REST APIs, Docker, and AWS.

DevOps Trainer (Internship)

Apr 2024 – Jul 2024

Bannari Amman Institute of Technology

Tamil Nadu, India

- Designed and delivered hands-on DevOps training to **50+ students** covering Linux, Git, Docker, Kubernetes, Jenkins, Terraform, Ansible, and AWS end-to-end.
- Demonstrated real-world **CI/CD pipeline** design and containerized deployment using Docker and Kubernetes on AWS infrastructure.
- Introduced system observability using **Prometheus and Grafana** for real-time performance monitoring, giving students production-grade exposure.

Projects

Health Insurance Claim Management System | [GitHub](#)

Dec 2024 – Apr 2025

- Java • Spring Boot • Spring Security • MySQL • Maven • Thymeleaf
- Led a team of 5** to build an end-to-end insurance claim management platform supporting **4 distinct user roles** — Admin, User, Agent, and Adjuster — each with dedicated dashboards and granular access controls.
 - Engineered **15+ RESTful API endpoints** with a complete Spring Security filter chain implementing **BCrypt password encryption**, session management, and URL-level authorization.
 - Improved **MySQL query performance by ~20%** through strategic indexing, optimized JPA queries, and Hibernate entity mapping, reducing claim retrieval latency.
 - Built a **document upload system** with UUID-based file naming and server-side storage, supporting MIME-type and file-size validation for claim evidence attachments.

- Implemented a **support ticket module** with priority-based lifecycle (OPEN → RESOLVED) for users to raise and track issues against active claims.
- Followed **clean architecture**: layered Controller → Service → Repository pattern with centralized exception handling and input validation throughout.

NOOL — Saree Polishing Workshop ERP |  GitHub |  Live

Apr 2025 – Present

- React 19 • Spring Boot • Spring Security • MySQL • JWT • Tailwind CSS
- Built originally to **replace 6 paper notebooks** used by a family-run saree-polishing workshop — eliminating 15–20 minutes of manual ledger lookups per query and reducing end-of-day calculation errors to zero.
 - Architected and delivered a **full-stack, role-based ERP system** covering inventory, payroll, attendance, and owner billing — with **4 distinct role front-ends** (Super Admin, Admin, Worker, Saree Owner) sharing a single stateless Spring Boot API.
 - Designed a **secure JWT authentication layer** (HS256, 24-hour expiry) with Spring Security enforcing role-based access control (RBAC) at both API endpoint and client-side route-guard levels.
 - Engineered an **immutable saree ledger** with pessimistic row-level locking to prevent concurrent receipt/return race conditions, ensuring inventory integrity across concurrent users.
 - Built **auto-calculated payroll and owner billing engines** — employee salary derived from daily polish counts × per-employee rate; owner invoices from quantity returned × per-owner rate — eliminating all manual arithmetic.
 - Developed **analytics dashboards and Excel export (xlsx)** for revenue trends, workforce productivity, and owner payment reconciliation; deployed React 19 SPA on **Vercel** — live at noolerp.vercel.app.

WordWise — Full Stack Text Analytics Platform |  GitHub

Apr 2024 – Jun 2024

- React.js • Spring Boot • AWS EC2 • MySQL RDS • Terraform • Tailwind CSS
- **Live-deployed** a full-stack text analytics app on AWS EC2 with real-time word, character, and space count analysis.
 - Designed **RESTful backend APIs** in Spring Boot integrated with a React.js frontend for real-time data processing with low-latency request-response cycles.
 - Provisioned **AWS infrastructure using Terraform (IaC)**: EC2 for compute, Amazon RDS (MySQL) for managed database, with security groups for HTTP/SSH/MySQL access.
 - Packaged and deployed the Spring Boot JAR on EC2 via SSH; served the React production build on port 80 — achieving a **fully automated, repeatable deployment process**.
 - Built a responsive, real-time UI with **React.js and Tailwind CSS** updating analytics on every keystroke for a seamless cross-device user experience.

Education

- B.E. – Electronics and Instrumentation Engineering
- Bannari Amman Institute of Technology
- CGPA: 7.49 / 10.0
- 2021 – 2025
- Tamil Nadu, India
- Higher Secondary Certificate (HSC)
- Saratha International Senior Secondary School
- Percentage: 78.4%
- May 2021
- Tamil Nadu, India

Certifications

GitHub Foundations	GitHub	 View Certificate
Java (Basic)	HackerRank	 View Certificate
Claude Certified Architect	Claude	 View Certificate

Extracurricular & Community

- Cloud Computing Trainer
- Bannari Amman Vidhya Niketan (School Outreach)
- Trained 50+ school students in Cloud Computing fundamentals using the AWS Console, covering virtual cloud concepts, EC2, S3, and IAM for hands-on understanding.
- Oct 2023
- Tamil Nadu, India
- Event Co-ordinator — Ersmernoz'23 & Ersmernoz'24
- Bannari Amman Institute of Technology
- Organized and co-ordinated the **paper presentation events** for two consecutive annual technical symposiums, managing scheduling, participant registration, and judging logistics.